ALGEBRA: ADVANCED CONCEPTS & SOLUTIONS

1. Quadratic Equations

Theory of Quadratic Roots

A quadratic equation $ax^2 + bx + c = 0$ has two roots given by:

$$x = [-b \pm \text{sqrt}(b^2 - 4ac)] / 2a$$

The Discriminant ($D = b^2 - 4ac$) defines the nature of roots:

- $D > 0$: Two distinct real roots.
- $D = 0$: Two identical real roots.
- $D < 0$: Two conjugate complex roots.

Sum of roots = $-b/a$, Product of roots = c/a.

2. Long Answer Problems: Quadratic Equations

Question: Solve $2x^2 - 7x + 3 = 0$ and verify the sum and product of roots.

> **Solution:**
> $D = (-7)^2 - 4(2)(3) = 49 - 24 = 25$.
> $x = [7 \pm 5] / 4$. Roots: $x = 12/4 = 3$ and $x = 2/4 = 0.5$.
> Sum = $3 + 0.5 = 3.5$. Formula: $-(-7)/2 = 3.5$ (Verified).
> Product = $3 * 0.5 = 1.5$. Formula: $3/2 = 1.5$ (Verified).

3. Polynomials & Remainder Theorem

Polynomial Foundations

Remainder Theorem: The remainder of polynomial $P(x)$ when divided by $(x - a)$ is $P(a)$.

Factor Theorem: If $P(a) = 0$, then $(x - a)$ is a factor of $P(x)$.

4. Partial Fractions

Decomposing rational expressions: $P(x)/Q(x) = A/(x-a) + B/(x-b)$.

Question: Resolve $5x / [(x+1)(x-2)]$ into partial fractions.

> **Solution:**
> $5x / [(x+1)(x-2)] = A/(x+1) + B/(x-2)$
> $5x = A(x-2) + B(x+1)$ If
> $x=2$: $10 = 3B \Rightarrow B =$
> $10/3$.
> If $x=-1$: $-5 = -3A \Rightarrow A = 5/3$.
> Result: $5/[3(x+1)] + 10/[3(x-2)]$.

5. Inequalities: Wavy Curve Method

Sign Scheme Analysis

Inequalities like $(x-a)(x-b) > 0$ are solved by identifying critical points on the number line and checking intervals.

Question: Solve $x^2 - 5x + 6 < 0$.

> **Solution:**
> $(x-2)(x-3) < 0$.
> Critical points are $x=2$ and $x=3$.
> Intervals: $(-\infty, 2)$, $(2, 3)$, $(3, \infty)$.
> Testing $x=2.5$: $(2.5-2)(2.5-3) = (0.5)(-0.5) = -0.25 < 0$.
> Solution: $x \in (2, 3)$.